

IDS2Eval Scorecard

cic-ids2018-fullscale
16 ok · 3 warning · 3 flag · 22 checks

REVIEW REQUIRED

- dedup_check

train duplicates: 0 (0.00%); test rows leaking a train feature-match: 0 (0.00%); test-internal duplicates: 0

ok
- label_conflict_check

no feature vector maps to more than one label

ok
- near_duplicate_class_check

no near-zero-distance feature vectors found across 15 classes tested

ok
- leakage_screen

no single feature or pair dominates importance (top1=7.1%, top2=12.7%)

ok
- one_rule_check

single rule 'Dst Port <= 121' reaches 88.8% test accuracy (88.6% train)

ok
- feature_auc_ranking_check

'Fwd Act Data Pkts' alone reaches AUC 1.000 identifying 'DDOS attack-LOIC-UDP'; a single feature this separable is worth checking for a leakage artifact

flag
- identity_column_flag

standalone AUC by column: {'Dst Port': 0.93}. Suggest dropping: ['Dst Port']

flag
- port_protocol_shortcut_check

standalone AUC of 'Dst Port' + 'Protocol' combined: 0.930. Port and protocol together may predict the label without attack behavior

flag
- temporal_leakage_check

no schema.timestamp_column configured

ok
- temporal_realism_check

no schema.timestamp_column configured

ok
- flow_group_leakage_check

no schema.flow_id_columns configured

ok
- row_order_leakage_check

adjacent-row label-transition rate (train=1.00, test=1.00) as a share of what a randomly shuffled ordering would produce

ok
- homogeneity_test

10 classes tested; test-to-train and train-internal proximity statistically indistinguishable for every class (consistent with inherent class homogeneity, not train/test leakage)

ok
- resplit_falsification

random-split accuracy=0.9823, grouped-split accuracy=0.9826 (drop=-0.0003); grouped split reproduces random-split accuracy, consistent with inherent class homogeneity rather than a split-artifact explanation

ok
- scenario_holdout_falsification

no schema.scenario_column configured

ok
- class_distribution_report

15 classes, train imbalance ratio (majority:minority) = 316063:1; classes below 1% of train: ['Infiltration', 'SSH-Bruteforce', 'DoS attacks-GoldenEye', 'DoS attacks-Slowloris', 'DDOS attack-LOIC-UDP', 'Brute Force -Web', 'Brute Force -XSS', 'SQL Injection', 'FTP-BruteForce', 'DoS attacks-SlowHTTPTest']

warning
- low_cardinality_warning

unique values by column: {'Dst Port': 64496}

ok
- schema_fingerprint_check

no known extractor signature matched

ok
- data_integrity_check

missing values in 1 feature column(s); +-inf values in 2 feature column(s); 8 constant/near-constant feature(s)

warning
- result_robustness_check

accuracy ranges from 0.9818 (identity_columns_dropped) to 0.9921 (deduplicated) across 4 conditions, a spread of 0.0104

ok
- known_issue_lookup

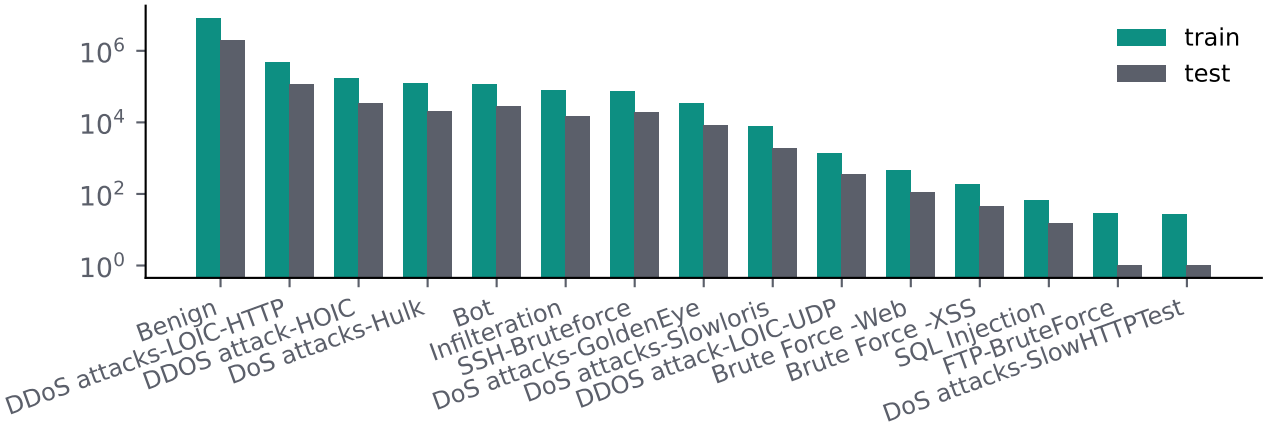
The 2018-02-23 day-file's Brute-Force-Web/Brute-Force-XSS rows are approximately 41% mislabeled (ground-truth labeling error, not an extraction or pipeline bug). (Liu et al. 2022, IEEE CNS); An independent re-labeling audit measured a 7.53% overall label corruption rate, with some individual attack classes above 75%. (Cantone et al. 2024, IEEE Access)

warning
- repeated_seed_falsification_check

grouped-split accuracy drop across 10 seeds: mean=+0.0000, 95% CI=[-0.0001, +0.0002]; not material: even the most generous estimate in the 95% CI (+0.0002) stays below the +0.0100 materiality threshold

ok

Class distribution (log scale)



Top feature importances (leakage_screen)

